

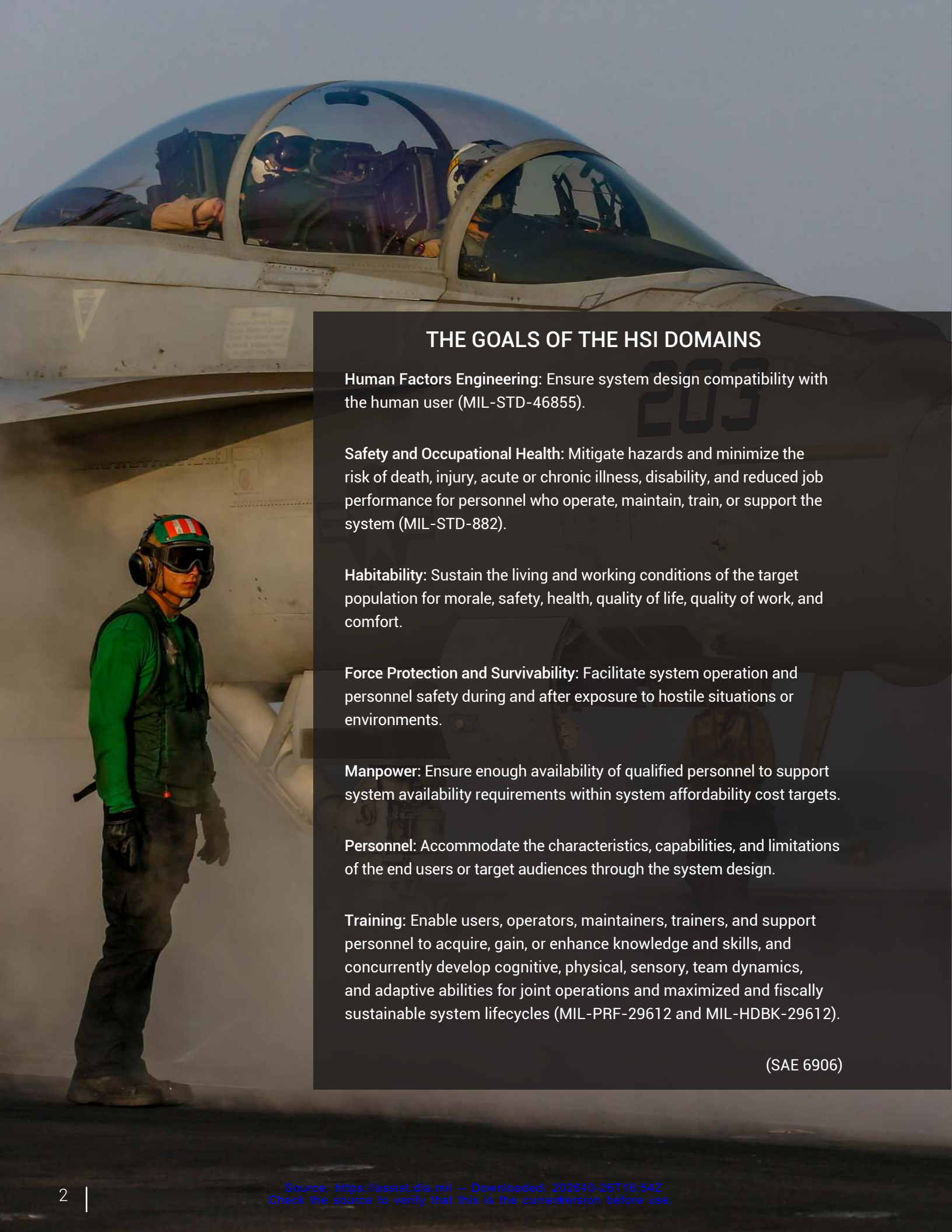
DEFENSE STANDARDIZATION PROGRAM

CASE STUDY

Standard Practice for Human System Integration

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THE GOALS OF THE HSI DOMAINS

Human Factors Engineering: Ensure system design compatibility with the human user (MIL-STD-46855).

Safety and Occupational Health: Mitigate hazards and minimize the risk of death, injury, acute or chronic illness, disability, and reduced job performance for personnel who operate, maintain, train, or support the system (MIL-STD-882).

Habitability: Sustain the living and working conditions of the target population for morale, safety, health, quality of life, quality of work, and comfort.

Force Protection and Survivability: Facilitate system operation and personnel safety during and after exposure to hostile situations or environments.

Manpower: Ensure enough availability of qualified personnel to support system availability requirements within system affordability cost targets.

Personnel: Accommodate the characteristics, capabilities, and limitations of the end users or target audiences through the system design.

Training: Enable users, operators, maintainers, trainers, and support personnel to acquire, gain, or enhance knowledge and skills, and concurrently develop cognitive, physical, sensory, team dynamics, and adaptive abilities for joint operations and maximized and fiscally sustainable system lifecycles (MIL-PRF-29612 and MIL-HDBK-29612).

(SAE 6906)

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Defense Standardization Program Case Study

Standard Practice for Human System Integration





BACKGROUND

Human system integration (HSI), a comprehensive management and technical approach, is applied to systems development and integration to optimize human performance, increase total system performance, and minimize total systems ownership costs (Society of Automotive Engineers International [SAE] 6906). As stated in the *Defense Acquisition Guidebook*, Chapter 5-3.2, the total system includes not only the prime mission equipment, but also the people who operate, maintain, and support the system. HSI addresses human considerations in the systems engineering process.

Recognizing the importance of HSI, DoD developed mandatory HSI direction for the program manager (PM) in DODD 5000.01, The Defense Acquisition System, and DODI 5000.02, *Operation of the Defense Acquisition System*. DODD 5000.01 requires the PM to "apply human systems integration to optimize total system performance (hardware, software, and human)." DODI 5000.02, Enclosure 7, states: "The Program Manager will plan for and implement HSI beginning early in the acquisition process and throughout the product life cycle. The goals will be to optimize total system performance and total ownership costs, while ensuring that the system is designed, operated, and maintained to effectively provide the user with the ability to complete their mission." DODI 5000.02 recommends integrating seven areas during HSI planning. Key to the success of HSI is the integration of these seven individual HSI domains:

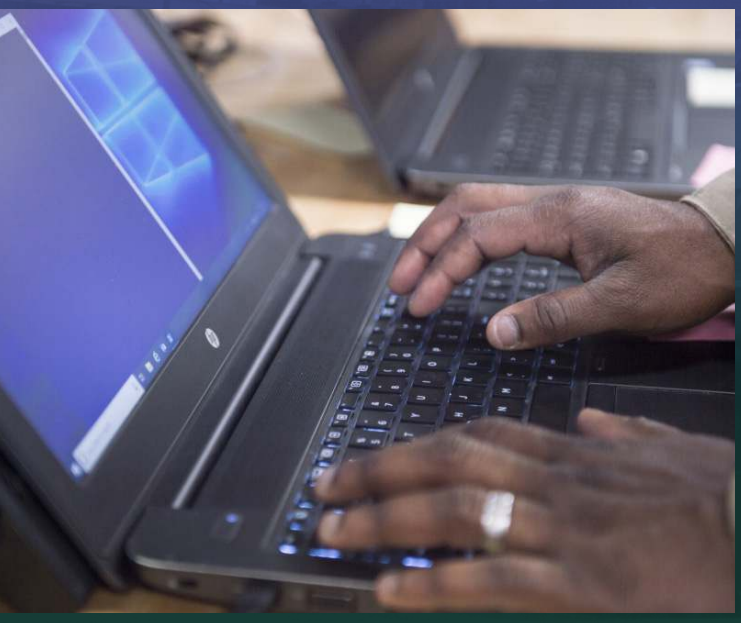
1. Human factors engineering
2. Safety and occupational health
3. Habitability
4. Force protection and survivability
5. Manpower
6. Personnel
7. Training.

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Many military and non-governmental standards (NGSs) address different aspects of HSI; however, no standards offered a comprehensive approach to HSI. The limitations of the standards included the following:

- The standard applies to a specific type of system, for example, ASTM F1337, which applies to maritime systems only.
- The standard addresses a specific HSI domain, such as MIL-STD-46855A, Human Engineering Requirements for Military Systems, Equipment, and Facilities, and MIL-STD-882E, Standard Practice for System Safety.
- The standard does not cover all HSI entirely, such as ISO 6385:2004, Ergonomic Principles in the Design of Work System, and Federal Aviation Administration (FAA) HF-STD-004, Requirements for a Human Factors Program.

Piecing together a disparate group of industry and military standards and guidebooks to define the HSI requirements is inefficient and could omit important HSI considerations for the warfighter.



As early as 2005, the services and industry began developing a standard DoD approach to HSI, when Office of the Secretary of Defense (OSD) leadership asked the DoD Human Factors Engineering Technical Advisory Group to create HSI requirements for industry. In June 2006, a DoD and industry workshop convened in response to the OSD request. At that time, the development of a standard proved too difficult. However, the G-45 HSI Committee (under the Government Electronics Industry Association then and currently under SAE International) undertook the development of a data item description for an HSI program plan, approved for use by DoD in 2007, and a data item description for an HSI report, approved for use by DoD in 2011. With these data item descriptions in place and cited in defense acquisition contracts, the timing favored the development of an HSI standard practice.

For contracts, a comprehensive standard needed to define the standard practice for procurement activities related to HSI and integrate the seven HSI domains. Naval Sea Systems Command (NAVSEA) organized an effort to develop a new, comprehensive HSI standard to furnish formal guidance for contractors to implement HSI on DoD system acquisition programs.

APPROACH

As the first step to address HSI standardization, NAVSEA established a DoD-level working group under the Joint Human Systems Integration Working Group (JHSIWG) chaired by OSD. NAVSEA subsequently created a charter for the HSI Standards Working Group (HSISWG) and obtained its approval from the JHSIWG and the Defense Standardization Council, solidifying support of leadership for the HSI standardization effort.

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The HSISWG performed a study and gap analysis of all the HSI standards and guidance. From the results, the HSISWG verified the need for a new standard. The HSISWG's first inclination was to develop a new military standard. However, at the urging of the Defense Standardization Program Office (DSPO), the group explored developing an NGS instead. The HSISWG sent a request for information (RFI) to four NGS bodies to evaluate the feasibility. The American National Standards Institute/Human Factors and Ergonomics Society, ASTM International, the Institute of Electrical and Electronics Engineers, and SAE received the RFI. The HSISWG evaluated their responses and chose SAE to develop the new HSI standard. SAE already had an active HSI Committee and had been involved in the development of data item descriptions for HSI. In addition, SAE envisioned evolving a portfolio of standards documents from the base HSI standard.

In May 2016, the G-45 HSI Committee began development of SAE 6906, Standard Practice for Human Systems Integration. HSI experts from industry, all branches of the military services, and U.S. and UK government agencies engaged in the development. In addition to DoD HSI experts, Navy representatives reached out to HSI experts at the Department of Homeland Security, the National Aeronautics and Space Administration, and FAA to help develop and review the content. The HSISWG created an outline with sections assigned to sub-teams to develop the content for the standard. By April 2017, the HSISWG assembled a first draft by compiling the content.

OUTCOME

The G-45 HSI Committee balloted and approved SAE 6906, Standard Practice for Human Systems Integration, in November 2018, with the standard published shortly after in 2019. DoD adoption of the standard is underway. Overall, the development and final approval of SAE standard involved assistance and feedback from experts from across DoD, industry, and G-45 HSI Committee members and liaisons. SAE 6906 furnishes a standard, comprehensive approach to HSI for inclusion as a requirement in DoD contracts.

BENEFITS

SAE 6906 consolidates the DoD requirements for supplying HSI support for system acquisition into a single document for the first time, including the following HSI components:

- Planning
- Execution
- Coordination (internal and external to DoD)
- Documentation
- Administration
- Quality control.

Why take the additional step to adopt a non-government standard?

Once an NGS is developed, DoD adoption offers the following advantages:

- Visibility through ASSIST, narrowing the search for an acceptable standard from broad to narrow for DoD users.
- Confidence in the standard, because it is technically adequate to meet the needs of the adopting activity.
- A DoD point of contact for questions or problems with the standard.
- Advice on how to apply it and other use guidance for DoD personnel.

(SD-9)

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The new standard addresses collaboration among HSI domains and coordination between HSI and other disciplines, resulting in optimized total system and human performance. In addition, the standard facilitates minimizing personnel-driven risks and customer ownership costs. Examples of personnel-driven risks in system design include designs reducing human productivity, mismatches of user manpower or skills with designs, and user interface problems. In addition, studies show that implementing a successful HSI program furnishes lifecycle savings of up to 40 times the amount invested in HSI early in the lifecycle of a program.

This SAE standard offers DoD a consistent approach to HSI and a documented standard for solicitations and contracts to communicate standard requirements for HSI in DoD system acquisition programs. In addition, the documentation of HSI requirements in a NGS accommodates a variety of customers, internal and external to DoD.

The G-45 HSI Committee developed and published SAE 6906 and maintains the standard. The committee focuses on processes, tools, requirements, and guidelines to ensure satisfactory HSI. Committee membership includes U.S. industry, DoD, foreign government, other U.S. federal, university, and foreign industry members as well as SAE staff members. Approximately 30% of the technical members are DoD personnel.

CURRENT STATUS AND FUTURE PLANS

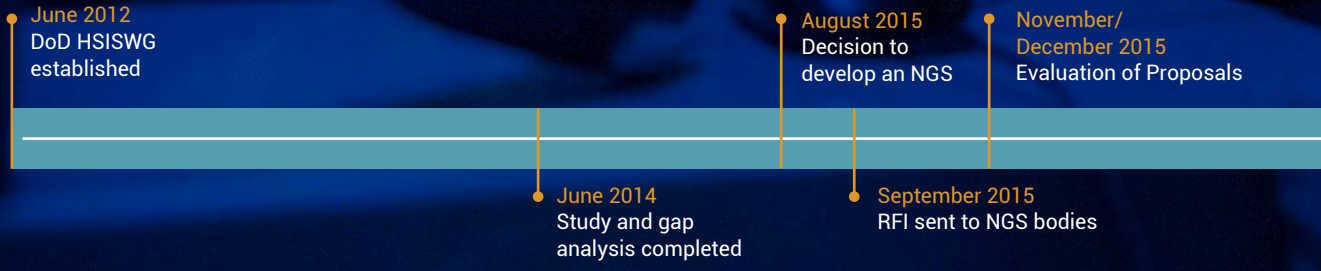
SAE 6906 was published in February 2019. An update to the standard, SAE 6906A, is underway, and will add an appendix to accommodate the DoD Adaptive Acquisition Framework. To facilitate implementation of SAE 6906 in the HSI community, the G-45 HSI Committee developed a 1-day classroom training course introducing HSI and the new standard (https://www.sae.org/learn/content/c1941/).

As a result of the HSISWG gap analysis, the group is creating HSI domain standard practices—documented as data item descriptions—for manpower, personnel, habitability, and force protection since no DoD-wide standard practices existed for these domains. The DoD HSISWG is also developing a military handbook with government guidance on how to use, apply, and tailor SAE 6906. The new handbook will address HSI responsibilities unique to the government and what the government must supply to enable industry to meet the government's HSI requirements.

KEYS TO SUCCESS

Support and coordination with stakeholders. The HSISWG stemmed from a meeting of the higher level JHSIWG and departmental standardization officers. The JHSIWG and Defense Standardization Council

Timeline for SAE 6906



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signed the charter for the new working group. Coordination with and support from these groups contributed to the success of the standard development, ensuring the project aligned with department policies on HSI and standards development. Early on, NAVSEA enlisted the support of HSI experts from DoD, industry, and other government agencies. Industry involvement was critical to creating the guidance that would be most valuable to helping industry understand and meet DoD HSI requirements.

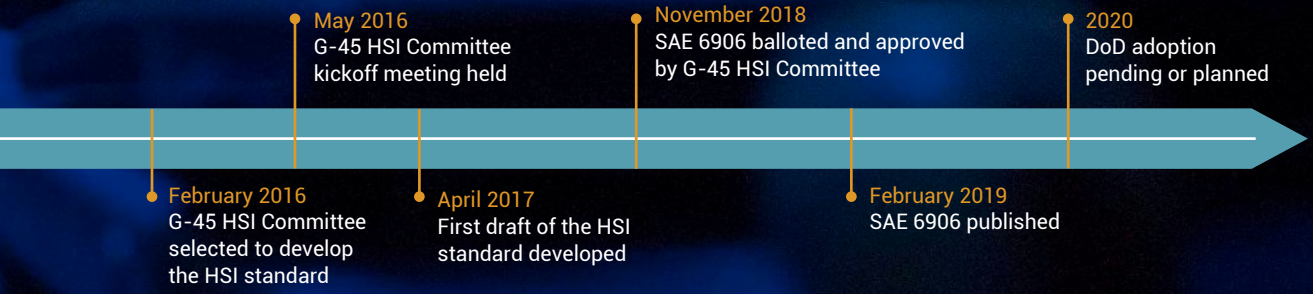
Gap identification and analysis of alternatives. The working group initially performed a gap analysis study to define the scope and select the necessary criteria for an HSI standard. The study reviewed all DoD and industry HSI standards to uncover discrepancies among the standards and evaluate whether any of the standards could meet DoD's requirement. The study outcome resulted in a clear definition of the needed standards for HSI and the various HSI domains. The study supplied the basis for the work to follow. The group then conducted an analysis of alternatives for closing the gaps found by the study. The analysis of alternatives recommended a solution and implementation approach.

Effective leadership. Effective leadership, passion, and perseverance fostered the successful development of SAE 6906. Three chairs led the 120-person committee with the support of government and industry vice chairs. Continuity



of leadership contributed to the effort. To keep the momentum going, the team met quarterly in person, held phone conferences between the in-person meetings, and established sub-teams to meet separately on assigned topics. They set aside time at the annual HSI conference to meet as well. Engaged members actively participated in the discussions. The committee chairs divided the work on the standard, appointing leads for the different sections to achieve the milestones. Leadership persevered through the multitude of tradeoffs and compromises, leading to a standard accepted by government and industry.

Partnering with an appropriate NGS body. The decision to develop an NGS, instead of a military standard, facilitated access to a diverse group of HSI experts from industry



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and government. These experts also furnished a variety of perspectives, including military customers with requirements for HSI and industry suppliers, who must satisfy the government's HSI requirements. The NGS body that the group selected, SAE, had extensive experience in human factors engineering and HSI standards development. The G-45 HSI Committee had already developed data item descriptions for an HSI program plan and an HSI report approved for use by DoD and cited in acquisition contracts. The team took advantage of the consensus standard development expertise and the associated tools available through its SAE partnership to track and manage the development of the standard.

LESSONS LEARNED

- Enlist sponsorship from leadership. The DoD HSIWG and the DSPO offered valuable support and guidance for the HSI standardization project.
- Conduct an analysis of gaps and alternatives as a first step to confirm the need for standardization and clearly define the project scope. Data from the analysis supported the case for the need for a standard with leadership. Agreeing on where the HSI process begins and ends was challenging, as well as deciding on what belonged in the overall process versus one of the HSI domains. This step furnished a shared focus for the project moving forward.
- Engage subject matter experts and stakeholders from industry and other government organizations early in the project. A team of HSI experts with varying perspectives from across government and industry prepared the standard.
- Work with an appropriate NGS body to develop a NGS when possible. The RFI and the decision process enabled the selection of an experienced body to work with the team for development and maintenance of the standard.
- Create an outline and aggressive timeline for drafting the standard and divide the workload for content development across sub-teams. The project lead established a 9-month deadline for the project and a regular meeting schedule to maintain momentum.

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